

```
% define training data
% note 1 = S, 2 = M, 3 = L
A=[55 160 1; 58 165 1; 60 160 2; 65 166 2; 70 168 2; 68 171 2; ...
  75 175 3; 75 180 3; 80 187 3; 90 190 3]
```

```
%plot data
```

```
x= A(:,1)
y= A(:,2)
c= A(:,3)
sz=100
%scatter(x,y,sz,c,'filled')
```

```
%define test data and k
```

```
b=[69 170]
```

```
k=3
```

```
%find distances using euclidean norm
```

```
difference = A(:, 1:2)-b
```

```
distance = sum(difference.^2,2)
```

```
%find labels of k nearest neighbors
```

```
[B,I]=mink(distance, k)
```

```
labels = A(I,3)
```

```
%find most common label among nearest neighbors
```

```
guess = mode(labels)
```

```
%now generate new plot with our guess for test data
```

```
b = [b guess]
A = [A;b]
```

```
x= A(:,1)
y= A(:,2)
c= A(:,3)
sz=100
scatter(x,y,sz,c,'filled')
```